

Sales Performance Dashboard Analysis

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Summary

The objective of this project was to analyze company sales data to identify revenue trends, top-performing products, regional performance, and profit margins. By processing raw data into an interactive Power BI dashboard, I aimed to provide actionable insights for business growth and cost optimization.

Part 1: Data Understanding and Cleaning

- **Data Integration:** Merged the primary dataset with the "Task 1" dataset to address gaps in information.
- **Merge Logic:** Utilized the Sales and Postal Code columns as the joining keys, as these shared identical value patterns across both files.

×

Merge

Select tables and matching columns to create a merged table.

train_original

	Product ID	Category	Sub-Category	Product Name	Sales	1
	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	
	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Round...	731.94	
	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Univer...	14.62	
	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	

SampleSuperstore_original

	State	Postal Code	2	Region	Category	Sub-Category	Sales	1	Quantity	Discount	Profit
	Kentucky	42420		South	Furniture	Bookcases	261.96		2	0	41.9136
	Kentucky	42420		South	Furniture	Chairs	731.94		3	0	219.582
	California	90036		West	Office Supplies	Labels	14.62		2	0	6.8714
ale	Florida	33311		South	Furniture	Tables	957.5775		5	0.45	-383.031

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

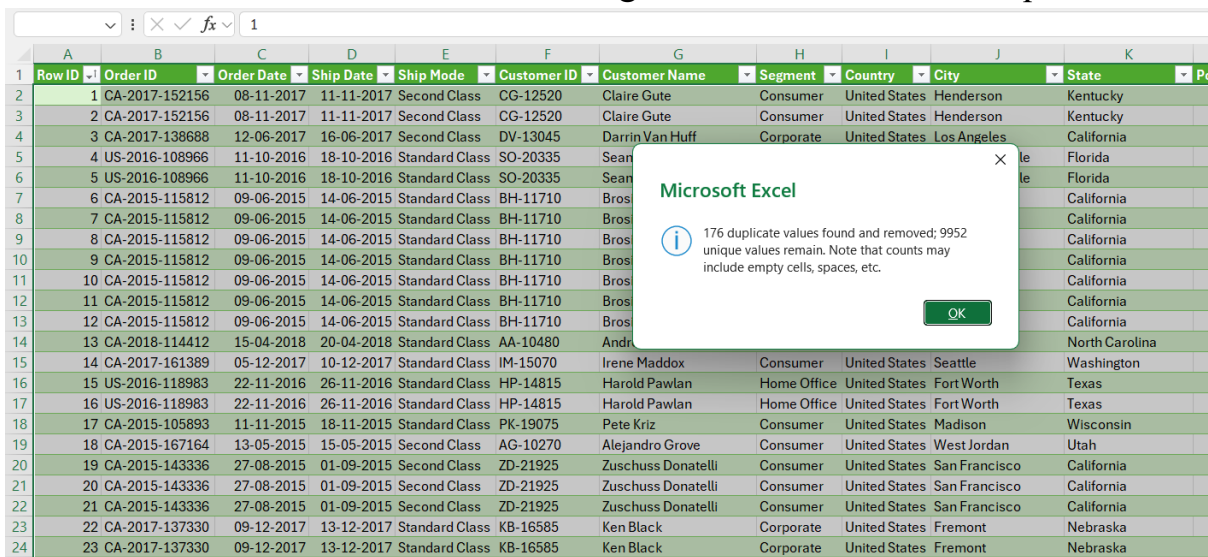
▷ Fuzzy matching options

✓ The selection matches 9789 of 9800 rows from the first table.

OK

Cancel

- **Duplicate Removal:** Performed a deduplication process to identify and remove redundant records, ensuring each transaction was unique.



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Product
1	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	
2	2	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky
3	3	CA-2017-138688	12-06-2017	16-06-2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California
4	4	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean			Florida	
5	5	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean			Florida	
6	6	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Bros			California	
7	7	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Bros			California	
8	8	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Bros			California	
9	9	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Bros			California	
10	10	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Bros			California	
11	11	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Bros			California	
12	12	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Bros			California	
13	13	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Bros			California	
14	14	CA-2018-114412	15-04-2018	20-04-2018	Standard Class	AA-10480	Andr			North Carolina	
15	15	CA-2017-161389	05-12-2017	10-12-2017	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington
16	16	US-2016-118983	22-11-2016	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas
17	17	US-2016-118983	22-11-2016	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas
18	18	CA-2015-105893	11-11-2015	18-11-2015	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin
19	19	CA-2015-167164	13-05-2015	15-05-2015	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah
20	20	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California
21	21	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California
22	22	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California
23	23	CA-2017-137330	09-12-2017	13-12-2017	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska
24	24	CA-2017-137330	09-12-2017	13-12-2017	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska

- **Missing Value Treatment:** Audited the combined dataset for null entries and handled missing values to prevent skewing the final KPI calculations.
- **Data Transformation:** Standardized data types (e.g., ensuring "Sales" is formatted as currency and "Order Date" as a date) to prepare for visual analysis in Power BI.

Part 2: KPI Analysis

The cleaned dataset was summarized using Excel Pivot Tables to calculate the aggregate performance across all transactions.

Key Findings

The analysis revealed the following high-level business results:

- **Total Sales:** \$2,271,859.43
- **Total Profit:** \$279,887.49
- **Profit Margin %:** 12%
- **Total Quantity Sold:** 37,627 units
- **Average Order Value (AOV):** \$228.28 per transaction

Significance

These KPIs provide a baseline for the business. A 12% profit margin suggests a healthy return, while an AOV of \$228.28 serves as a benchmark for future sales strategies aimed at increasing transaction size.

2271859.43469995						
A	B	C	D	E	F	
1						
2						
3	Sum of Sales	Sum of SampleSuperstore_original.Profit	Sum of SampleSuperstore_original.Quantity	Count of Order ID	Average Order Value (AOV)	Profit Margin
4	\$22,71,859.43	\$2,79,887.49	37627	9952	\$228.28	12%
5						
6						
7						
8						
9						
10						

Part 3: Business Analysis

The goal of this part was to take a closer look at the company's sales and profit data from different angles—like products, locations, and time.

It helps identify the top 10 products, products that are causing losses, and the best-performing regions. These insights make it easier to take smarter, data-driven decisions.

1. Which are the **Top 10 products** by sales?

- **Market Leader:** The **Canon imageCLASS 2200 Advanced Copier** is the top-performing product, generating **\$61,599.82** in revenue.
- **Key Contributors:** Other significant revenue drivers include the **Fellowes PB500 Electric Punch** (\$27,453.38) and the **Cisco TelePresence System EX90** (\$22,638.48).
- **Aggregate Value:** The top 10 products alone account for a grand total of **\$244,620.20** in sales.

	A	B
1	Top 10 Products by Sale	
2		
3	Row Labels	Sum of Sales
4	Canon imageCLASS 2200 Advanced Copier	\$61,599.82
5	Cisco TelePresence System EX90 Videoconferencing Unit	\$22,638.48
6	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	\$27,453.38
7	GBC DocuBind P400 Electric Binding System	\$17,965.07
8	GBC DocuBind TL300 Electric Binding System	\$19,823.48
9	GBC Ibimaster 500 Manual ProClick Binding System	\$19,024.50
10	Hewlett Packard LaserJet 3310 Copier	\$18,839.69
11	High Speed Automatic Electric Letter Opener	\$17,030.31
12	HON 5400 Series Task Chairs for Big and Tall	\$21,870.58
13	HP Designjet T520 Inkjet Large Format Printer - 24" Color	\$18,374.90
14	Grand Total	\$2,44,620.20
15		

2. Which products are making losses?

- **Highest Loss:** The **Cubify CubeX 3D Printer Double Head Print** is the most significant loss-maker, with a deficit of - **\$8,879.97**.
- **Persistent Issues:** Other high-tech items like the **Lexmark MX611dhe Monochrome Laser Printer** (-\$4,589.97) and the **Cubify CubeX 3D Printer Triple Head Print** (-\$3,839.99) are also contributing heavily to profit erosion.
- **Category Trend:** A notable trend is visible among furniture items, specifically **Conference Tables** (e.g., Chromcraft and Bush Advantage models), which appear frequently in the loss-making list with deficits ranging from **-\$2,500 to -\$2,700**.

	A	B
1	Products Making Losses	
2		
3	Row Labels	Sum of SampleSuperstore_original.Profit
4	Cubify CubeX 3D Printer Double Head Print	-\$8,879.97
5	Lexmark MX611dhe Monochrome Laser Printer	-\$4,589.97
6	Cubify CubeX 3D Printer Triple Head Print	-\$3,839.99
7	Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	-\$2,732.86
8	Bush Advantage Collection Racetrack Conference Table	-\$2,545.26
9	GBC DocuBind P400 Electric Binding System	-\$1,878.17
10	Cisco TelePresence System EX90 Videoconferencing Unit	-\$1,811.08
11	Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	-\$1,682.67
12	Martin Yale Chadless Opener Electric Letter Opener	-\$1,299.18
13	Balt Solid Wood Round Tables	-\$1,201.06
14	BoxOffice By Design Rectangular and Half-Moon Meeting Room Tables	-\$1,148.44
15	Riverside Furniture Oval Coffee Table, Oval End Table, End Table with Drawer	-\$1,147.40
16	Epson TM-T88V Direct Thermal Printer - Monochrome - Desktop	-\$1,057.23
17	Hon 2090 "Pillow Soft" Series Mid Back Swivel/Tilt Chairs	-\$989.05
18	O'Sullivan 4-Shelf Bookcase in Odessa Pine	-\$975.10
19	Bretford "Just In Time" Height-Adjustable Multi-Task Work Tables	-\$964.19
20	Zebra GK420t Direct Thermal/Thermal Transfer Printer	-\$938.28
21	3.6 Cubic Foot Counter Height Office Refrigerator	-\$872.08
22	Bevis Oval Conference Table, Walnut	-\$856.01
23	Tennisco Single-Tier Lockers	-\$825.75
24	Balt Conference Table	-\$705.07

3. Which region/state/city has the highest sales?

- **Regional Dominance:** The **Central** region shows strong performance with total sales of **\$4,92,879.99**.
- **Top Performing State:** Within the Central region, **Illinois** is a major driver, contributing **\$79,280.20** to the total.
- **City-Level Insights:** **Detroit** (Michigan) is a standout city with high sales of **\$42,446.94**.
- **Market Variance:** There is a significant gap between high-performing hubs like **Detroit** and emerging or lower-volume areas like **Garden City** (\$312.38) or **Oswego** (\$321.98), indicating varying levels of market demand or sales focus.

	A	B	C	D	E	F	G	H	I
1	Region/City/State wise Highest and Lowest Sale								
2									
3	Row Labels	Sum of Sales							
4	▢ Central	\$4,92,879.99							
5	▢ Illinois	\$79,280.20							
6	▢ Indiana	\$48,718.40							
7	▢ Iowa	\$4,443.56							
8	▢ Kansas	\$2,914.31							
9	Olathe	\$895.79							
10	Wichita	\$825.27							
11	Overland Park	\$606.91							
12	Garden City	\$312.38							
13	Manhattan	\$273.96							
14	▢ Michigan	\$76,163.87							
15	Detroit	\$42,446.94							
16	Jackson	\$15,447.84							
17	Midland	\$5,291.51							
18	Westland	\$2,057.98							
19	Trenton	\$1,765.20							
20	Lansing	\$1,610.29							
21	Dearborn	\$1,603.31							

4. Which region/state/city has the lowest sales?

- **Lowest Performing City: Abilene** (Texas) recorded the absolute lowest sales figure in the dataset at **\$1.39**.
- **Bottom Performers:** Several other cities showed negligible revenue, including **Elyria** (\$1.82), **Jupiter** (\$2.06), and **Pensacola** (\$2.21).
- **State-Level Gaps:** While states like Illinois and Michigan drive high volumes, cities in states such as **Texas, Florida, and Ohio** appear frequently in the bottom 10 list, suggesting inconsistent performance within those states.
- **Regional Underperformance:** Areas such as **Manhattan** (Kansas) in the Central region represent the lower bound of regional contribution, with sales of only **\$273.96**.

5. Which category/sub-category contributes the most profit?

- **Top Profitable Category: Technology** is the primary profit driver, contributing a total of **\$1,44,206.51**. Within this category, **Copiers** (\$55,320.83) and **Phones** (\$44,287.60) are the standout sub-categories.
- **Steady Contributor: Office Supplies** follows with a strong profit contribution of **\$1,19,855.53**, led largely by **Paper** (\$33,763.29) and **Binders** (\$29,684.15).

- **Underperforming Category: Furniture** is the least profitable category, yielding only **\$15,825.45** despite having high-sales sub-categories like **Chairs** (\$25,796.16).
- **Critical Profit Leakage:** The **Tables** sub-category within Furniture is the most significant outlier, resulting in a substantial loss of **-\$18,223.93**. **Bookcases** also show a deficit of **-\$4,377.82**, further eroding Furniture's overall margins.

	A	B	C	D	E	F	G
1	Category and Sub-category contributing to Profit						
2							
3	Row Labels	Sum of SampleSuperstore_original.Profit					
4	▢ Furniture	\$15,825.45					
5	Chairs	\$25,796.16					
6	Furnishings	\$12,631.04					
7	Bookcases	-\$4,377.82					
8	Tables	-\$18,223.93					
9	▢ Office Supplies	\$1,19,855.53					
10	Paper	\$33,763.29					
11	Binders	\$29,684.15					
12	Storage	\$20,697.68					
13	Appliances	\$17,063.59					
14	Envelopes	\$6,877.74					
15	Art	\$6,528.18					
16	Labels	\$5,552.51					
17	Fasteners	\$947.33					
18	Supplies	-\$1,258.94					
19	▢ Technology	\$1,44,206.51					
20	Copiers	\$55,320.83					
21	Phones	\$44,287.60					
22	Accessories	\$41,213.32					
23	Machines	\$3,384.76					
24	Grand Total	\$2,79,887.49					
25							

6. Which months have the highest and lowest sales?

- **Year-over-Year Growth:** The company experienced a dramatic surge in performance, with sales climbing from approximately **\$481K in 2015** to over **\$733K by 2018** (as seen in your Power BI trend line).
- **Peak Sales Months:** Across the years, **September and November** consistently emerge as high-volume months. For example, in 2015, September reached **\$81,644.95** and November reached **\$78,041.44**.
- **Low Sales Periods:** **February** is consistently the lowest performing month of the year, with 2015 recording only **\$4,666.81** in sales.

7. Identify seasonal trends in sales.

- **Seasonal Surge:** There is a clear "End-of-Year" surge starting in September and peaking through December, likely driven by holiday shopping or annual corporate budget clearances.

Part 4: Dashboard Creation

To conclude the report, I created a final interactive dashboard using Power BI:

- **Interactive Dashboard:** Built in Power BI, featuring dynamic filters for **Year, Region, and Category** to allow for real-time data exploration.
- **Visualizations Included:**
 - **Sales Trend Line:** Highlighting the multi-year growth trajectory.
 - **Geographic Map:** Displaying sales density across the United States.
 - **Profitability Bar Chart:** Clearly identifying the "Tables" loss-making outlier.

